

# **Microelectronics**

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# 1 Microelectronics

## I Signal

### i Signal

A signal is a function that conveys information about a phenomenon. Signal processing is usually most conveniently performed by electronic systems on voltage or current signals, especially the former. Only voltage and current signals are studied here.

### ii Signal source

A signal source is typically represented in the Thévenin form, a voltage source in a series connection with a resistance, or the Norton form, a current source in a parallel connection with a resistance.

### iii Cascade

Output of a component as the input of another.

### iv Transducer

A transducer is a device that convert a signal to a voltage or current signal or vice versa.

### v Signal processing

Signal processing includes analog signal processing (ASP), analog-to-digital converter (ADC), digital signal processor (DSP), digital-to-analog converter (DAC), amplifier, etc.

### vi Coupling capacitor

A capacitor connected in series between stages to pass AC signal while blocking DC.

### vii Symbol convention

- Lowercase character with uppercase subscript means total signal, e.g.  $i_C(t)$ .
- Lowercase character with lowercase subscript means AC signal, e.g.  $i_c(t)$ .
- Uppercase character with uppercase subscript means DC signal, e.g.  $I_C$ .
- Uppercase character with lowercase subscript means peak signal, e.g.  $I_c$ .

Their relation is

$$i_C(t) = i_c(t) + I_C$$

$$i_c(t) = I_c \sin(\omega t + \varphi)$$

### viii differential and common-mode signal

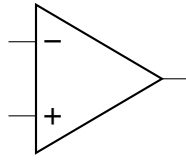
For a component with two input signals, the differential signal (AC signal)  $v_d$  is the difference between the two input signals, and the common-mode signal (DC signal)  $v_{cm}$  is the algebraic average of the two input signals.

## II Amplifier (amp)

### i Amplifier

An electronic device which amplifies an electrical signal by a scale called amplifier gain  $A$  and optionally convert voltage signal to current signal or vice versa.

### ii Diagram



### iii Two-port model

An amplifier can be modeled as a two port model, where an one input port and one output port is related by the amplifier, which acts as a controlled source controlled by the input port voltage or current which enforces voltage or current on the output port.

### iv Circuit model of voltage amplifier

A voltage amplifier draws current from a source  $v_s$  with internal resistance  $R_s$  with input resistance  $R_i$  and voltage  $v_{in}$  across it and supplies  $v_c = \mu v_{in}$  to a load network with output resistance  $R_o$  and load  $R_L$ , where  $\mu$ , also denoted  $A_{vo}$ , is open-loop voltage gain factor.

$$\frac{v_{in}}{v_s} = \frac{R_i}{R_s + R_i}$$

$$\frac{v_c}{v_{in}} = \mu$$

$$\frac{v_{out}}{v_c} = \frac{R_L}{R_o + R_L}$$

Voltage gain:

$$A_v = \frac{v_{out}}{v_{in}} = \mu \frac{R_i R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\mu|$  due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \ll R_L, \quad A_v \approx \mu$$

## v Circuit model of current amplifier

A current amplifier draws current from a current source  $i_s$  with internal resistance  $R_s$  in parallel with input resistance  $R_i$  and current  $i_{in}$  through it and supplies  $i_c = \beta i_{in}$  by CCCS to a load network with output resistance  $R_o$  in parallel and load  $R_L$ , where  $\beta$ , also denoted  $A_{cs}$ , is short-circuit current gain factor.

$$\begin{aligned}\frac{i_{in}}{i_s} &= \frac{R_s}{R_s + R_i} \\ \frac{i_c}{i_{in}} &= \beta \\ \frac{i_{out}}{i_c} &= \frac{R_o}{R_o + R_L}\end{aligned}$$

Current gain:

$$A_c = \frac{i_{out}}{i_{in}} = \beta \frac{R_s R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\beta|$  due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \gg R_L, \quad A_c \approx \beta$$

## vi Circuit model of transconductance amplifier

A transconductance amplifier draws current from a voltage source  $v_s$  with internal resistance  $R_s$  in series with input resistance  $R_i$  and current  $i_{in}$  through it and supplies  $i_c = g v_{in}$  by VCCS to a load network with output resistance  $R_o$  in parallel and load  $R_L$ , where  $\mu$  is open-loop transconductance factor.

$$\begin{aligned}\frac{v_{in}}{v_s} &= \frac{R_i}{R_s + R_i} \\ \frac{i_c}{v_{in}} &= g \\ \frac{i_{out}}{i_c} &= \frac{R_o}{R_o + R_L}\end{aligned}$$

Gain:

$$A_g = \frac{i_{out}}{v_{in}} = g \frac{R_i R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |g|$  due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \gg R_L, \quad A_g \approx g$$

## vii Circuit model of transresistance amplifier

A transresistance amplifier draws current from a current source  $i_s$  with internal resistance  $R_s$  in parallel with input resistance  $R_i$  and current  $i_{in}$  through it and supplies  $v_c = r i_{in}$  by CCVS to a load network with output resistance  $R_o$  in series and load  $R_L$ , where  $r$  is short-circuit transresistance factor.

$$\frac{i_{in}}{i_s} = \frac{R_s}{R_s + R_i}$$

$$\frac{v_c}{i_{in}} = r$$

$$\frac{v_{out}}{v_c} = \frac{R_L}{R_o + R_L}$$

Gain:

$$A_r = \frac{v_{out}}{i_{in}} = \beta \frac{R_s R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |r|$  due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \ll R_L, \quad A_c \approx r$$

### viii Amplifier gain in decibels (dB)

A power gain  $A_p$  in dB is defined as

$$G_p = 10 \log_{10}(A_p)$$

A voltage gain  $A_v$  in dB is defined as

$$G_v = 20 \log_{10}(A_v)$$

A current gain  $A_c$  in dB is defined as

$$G_c = 20 \log_{10}(A_c)$$

### ix Saturation

The linear characteristic only remains within a range. A positive and negative saturation levels can be observed.

### x S parameter

If the energy enters through port  $i$  and emerges from port  $j$ , the S parameter is named  $S_{ji}$ .

### xi Unilateral amplifier

An unilateral amplifier is an amplifier in which the signal transfer is effectively one-way, that is, the output signal is only dependent on the input signal but not what is connected to the output terminal. That is for each output port  $j$  and each input port  $i$ ,  $S_{ij} = 0$ .

### xii Power supply

To amplify signal, power supplies are needed, but often omitted in the diagram.

- $V_{CC+}$  or  $V_{CC}$ : positive supply voltage, denoted on the + side of an op-amp or omitted;
- $V_{CC-}$  or  $-V_{EE}$ : negative supply voltage, denoted on the – side of an op-amp or omitted.

The DC power delivered is the sum of power from the positive and negative power supply terminals  $V_{CC}$  and  $V_{EE}$

$$P_{DC} = V_{CC}I_{CC} + V_{EE}I_{EE}$$

Suppose input and output signals have power  $P_I$  and  $P_L$ .

$$P_{DC} + P_I = P_L + P_{\text{dissipated}}$$

where  $P_{\text{dissipated}}$  is the power dissipated.

### xiii Classification based on frequency response

A band-pass amplifier is also called tuned amplifier. A low-pass amplifier is also called DC amplifier. A amplifier directly coupled with a coupling capacitor, called capacitively coupled, is called AC amplifier.

## III Operational Amplifier (op amp or op-amp)

### i Definition

An op amp has five terminals:

- $v_+$ : non-inverting input voltage of the terminal at the + sign;
- $v_-$ : inverting input voltage of the terminal at the – sign;
- $v_o$ : output voltage of the terminal at the opposite side of inputs;
- $V_{CC+}$  or  $V_{CC}$ : positive supply voltage, denoted on the + side of an op-amp or omitted;
- $V_{CC-}$  or  $-V_{EE}$ : negative supply voltage, denoted on the – side of an op-amp or omitted.

$$v_o = A(v_+ - v_- - v_{os})$$

where  $A$  is open-loop gain or open-circuit gain, and  $v_{os}$  is input offset voltage.

$$v_{ld} = v_+ - v_-$$

$$v_{lcm} = \frac{v_+ + v_-}{2}$$

### ii Ideal op amp

- Infinite input impedance: No signal current flowing into  $v_+$  and  $v_-$ .
- Infinite open-loop gain
- Infinite bandwidth: open-loop gain remains constant in all frequencies.
- Zero output impedance: The amplifier is unilateral.
- Zero input offset voltage:  $v_{os} = 0$ , which implies zero common-mode gain, i.e., infinite common-mode rejection.

This is a linear element with inputs  $v_+$  and  $v_-$  and output  $v_o$ .

We can construct a virtual short circuit between  $v_+$  and  $v_-$  for ideal op amp.

Note that while  $v_+ - v_-$ ,  $v_o$  may be any voltage since  $A = \infty$  and  $0 * \infty$  can be any number in  $[0, \infty]$  in limit sense.

### iii Feedback configuration

To have stable signal, op amp in a circuit is connected with a negative feedback configuration, which consists of the op amp and a connection from  $v_o$  to either of the input terminals with a resistance on it, called feedback connection. When the feedback connection connects to  $v_-$ , it is called inverting configuration. When the feedback connection connects to  $v_+$ , it is called non-inverting configuration.

### iv Inverting amplifier

An inverting amplifier consists of an op amp where input voltage  $v_i$  is connected to  $v_-$  through an input resistance  $R_i$  (current from  $v_i$  to  $v_-$ ),  $v_+$  is connected to ground, and  $v_o$  is fed back to  $v_-$  through a feedback resistance  $R_f$  (current from  $v_-$  to  $v_o$ ).

$$\begin{aligned} v_+ &= v_- = 0 \\ i_i &= \frac{v_i}{R_i} = i_f = -\frac{v_o}{R_f} \\ v_o &= -\frac{R_f}{R_i} v_i \end{aligned}$$

The closed-loop gain  $G$  is defined as

$$G = \frac{v_o}{v_i} = -\frac{R_f}{R_i}$$

When  $-G < 1$ , it's called an attenuator.

Effect of finite open-loop gain:

$$\begin{aligned} v_+ &= 0 \\ v_o &= -A v_- \\ i_i &= \frac{v_i - v_-}{R_i} = \frac{v_i + \frac{v_o}{A}}{R_i} = i_f = -\frac{v_o + \frac{v_o}{A}}{R_f} \\ G &= \frac{v_o}{v_i} = -\frac{R_f}{R_i \left( 1 + \frac{1 + \frac{R_f}{R_i}}{A} \right)} = \frac{G_{\text{ideal}}}{1 + \frac{1 - G_{\text{ideal}}}{A}} \\ \lim_{A \rightarrow \infty} G &= -\frac{R_f}{R_i} = G_{\text{ideal}} \end{aligned}$$

## v Transresistance amplifier

A transresistance amplifier consists of an op amp where input current  $i_i$  is connected to  $v_-$ ,  $v_+$  is connected to ground, and  $v_o$  is fed back to  $v_-$  through two resistances  $R_o$  (current from  $x$  to  $v_o$ ) and  $R_f$  (current from  $v_-$  to  $x$ ) in series connection, with the node  $x$  between  $R_o$  and  $R_f$  connected to ground through a resistance  $R_g$  (current from  $x$  to ground).

$$v_x = -R_f i_i$$

$$i_g = -\frac{R_f}{R_g} i_i$$

$$i_o = \left(1 + \frac{R_f}{R_g}\right) i_i$$

$$v_o = -R_f i_i - R_o i_o = -\left(R_f + R_o + \frac{R_f R_o}{R_g}\right) i_i$$

$$\frac{v_o}{i_i} = -\left(R_f + R_o + \frac{R_f R_o}{R_g}\right)$$

## vi Weighted summing amplifier

A weighted summing amplifier consists of an op amp, input voltages  $v_1, v_2, \dots, v_n$ ,  $n$  input resistors  $R_1, R_2, \dots, R_n$ , and a feedback resistor  $R_f$ , where for any  $k \in \mathbb{N} \wedge k \leq n$ ,  $v_k$  is connected to  $v_-$  through  $R_k$ ,  $v_+$  is connected to ground, and  $v_o$  is fed back to  $v_-$  through  $R_f$ .

$$i_k = \frac{v_k}{R_k}$$

$$\sum_{k=1}^n i_k = i_f = \frac{-v_o}{R_f}$$

$$v_o = -R_f \sum_{k=1}^n \frac{v_k}{R_k}$$

When  $R_k = R$  for all  $k \in \mathbb{N} \wedge k \leq n$ :

$$v_o = -\frac{R_f}{R} \sum_{k=1}^n v_k$$

## vii Non-inverting amplifier

A non-inverting (or noninverting) amplifier consists of an op amp where input voltage  $v_i$  is connected to  $v_+$ ,  $v_-$  is connected ground through a ground resistance  $R_g$  (current from  $v_-$  to ground), and  $v_o$  is fed back to  $v_-$  through a feedback resistance  $R_f$  (current from  $v_o$  to  $v_-$ ).

$$v_+ = v_- = v_i$$

$$i_g = \frac{v_i}{R_g} = i_f = \frac{v_o - v_i}{R_f}$$

$$v_o = \left(1 + \frac{R_f}{R_g}\right) v_i$$

The closed-loop gain  $G$  is defined as

$$G = \frac{v_o}{v_i} = 1 + \frac{R_f}{R_g}$$

Effect of finite open-loop gain:

$$\begin{aligned} v_+ &= v_i \\ v_o &= A(v_i - v_-) \\ v_- &= v_i - \frac{v_o}{A} \\ i_g = \frac{v_-}{R_g} &= \frac{v_i - \frac{v_o}{A}}{R_g} = i_f = \frac{v_o - v_-}{R_f} = \frac{v_o - v_i + \frac{v_o}{A}}{R_f} \\ G = \frac{v_o}{v_i} &= \frac{1 + \frac{R_f}{R_g}}{1 + \frac{1 + \frac{R_f}{R_g}}{A}} = \frac{G_{\text{ideal}}}{1 + \frac{G_{\text{ideal}}}{A}} \\ \lim_{A \rightarrow \infty} G &= 1 + \frac{R_f}{R_g} = G_{\text{ideal}} \end{aligned}$$

### viii Unity gain amplifier, unity gain buffer, buffer amplifier, or voltage follower

A unity gain amplifier consist of an op amp where input voltage  $v_i$  is connected to  $v_+$ , and  $v_o$  is fed back to  $v_-$ .

$$v_o = v_- = v_+ = v_i$$

### ix Difference amplifier or differential amplifier

A difference amplifier consists of an op amp where first input voltage  $v_1$  is connected to  $v_-$  through an input resistance  $R_1$  (current from  $v_1$  to  $v_-$ ), second input voltage  $v_2$  is connected to  $v_+$  through an input resistance  $R_2$  (current from  $v_2$  to  $v_+$ ),  $v_+$  is connected to ground through a ground resistance  $R_g$  (current from  $v_+$  to ground), and  $v_o$  is fed back to  $v_-$  through a feedback resistance  $R_f$  (current from  $v_-$  to  $v_o$ ). Denote common-mode signal  $v_{cm} = \frac{v_1 + v_2}{2}$  and differential signal  $v_d = v_2 - v_1$ .

$$\begin{aligned} v_o &= -\frac{R_f}{R_1}v_1 + \frac{R_g}{R_2 + R_g} \left( 1 + \frac{R_f}{R_1} \right) v_2 \\ i_1 &= \frac{1}{R_1} \left( v_1 - \frac{R_g}{R_2 + R_g} v_2 \right) \\ i_2 &= \frac{v_2}{R_2 + R_g} \end{aligned}$$

Common-mode gain  $A_{cm}$  and differential gain  $A_d$ , defined by

$$v_o = v_{cm}A_{cm} + v_dA_d,$$

are

$$A_{cm} = \frac{R_g}{R_2 + R_g} \left( 1 - \frac{R_2}{R_g} \frac{R_f}{R_1} \right)$$

$$A_d = \frac{R_g}{R_2 + R_g} \left( 1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1} \right)$$

Common-mode rejection ratio (CMRR) is defined as

$$\text{CMRR} = 20 \log \left| \frac{A_d}{A_{cm}} \right|$$

To make  $A_{cm} = 0$ , we need

$$\frac{R_f}{R_1} = \frac{R_g}{R_2},$$

at which

$$A_d = \frac{R_f}{R_1} = \frac{R_g}{R_2}$$

Differential input resistance  $R_d$ , defined as, when we connect  $v_1$  and  $v_2$  to a source  $v_d$  ( $v_d = v_2 - v_1$  and  $i_d = i_2 = -i_1$ ),

$$\frac{v_d}{i_d},$$

is

$$R_d = R_1 + R_2$$

Thus, high differential gain hurts differential input resistance, which lower the voltage applied to the amplifier since source has internal resistance.

*Proof.* Use superposition. With  $v_1$  enabled and  $v_2$  disabled,

$$v_+ = v_- = 0$$

$$i_2 = i_g = 0$$

$$i_1 = \frac{v_1}{R_1} = i_f$$

$$v_o = -i_f R_f = -\frac{R_f}{R_1} v_1$$

With  $v_2$  enabled and  $v_1$  disabled,

$$i_2 = i_g = \frac{v_2}{R_2 + R_g}$$

$$v_+ = v_- = i_g R_g = \frac{R_g}{R_2 + R_g} v_2$$

$$i_1 = -\frac{v_-}{R_1} = -\frac{R_g}{R_1(R_2 + R_g)} v_2 = i_f$$

$$v_o = v_- - i_f R_f = \frac{R_g}{R_2 + R_g} v_2 + \frac{R_g R_f}{R_1(R_2 + R_g)} v_2 = \frac{R_g}{R_2 + R_g} \left( 1 + \frac{R_f}{R_1} \right) v_2$$

Sum:

$$v_o = -\frac{R_f}{R_1} v_1 + \frac{R_g}{R_2 + R_g} \left( 1 + \frac{R_f}{R_1} \right) v_2$$

$$i_1 = \frac{v_1}{R_1} - \frac{R_g}{R_1(R_2 + R_g)} v_2 = \frac{1}{R_1} \left( v_1 - \frac{R_g}{R_2 + R_g} v_2 \right)$$

$$i_2 = \frac{v_2}{R_2 + R_g}$$

$$v_1 = v_{cm} - \frac{v_d}{2}$$

$$v_2 = v_{cm} + \frac{v_d}{2}$$

$$\begin{aligned} v_o &= -\frac{R_f}{R_1} \left( v_{cm} - \frac{v_d}{2} \right) + \frac{R_g}{R_2 + R_g} \left( 1 + \frac{R_f}{R_1} \right) \left( v_{cm} + \frac{v_d}{2} \right) \\ &= \frac{R_g}{R_2 + R_g} \left( 1 - \left( \frac{R_2 + R_g}{R_g} - 1 \right) \frac{R_f}{R_1} \right) v_{cm} + \frac{R_g}{R_2 + R_g} \left( 1 + \left( \frac{R_2 + R_g}{R_g} + 1 \right) \frac{R_f}{R_1} \right) v_d \\ &= \frac{R_g}{R_2 + R_g} \left( 1 - \frac{R_2 R_f}{R_g R_1} \right) v_{cm} + \frac{R_g}{R_2 + R_g} \left( 1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1} \right) v_d \end{aligned}$$

$$A_{cm} = \frac{R_g}{R_2 + R_g} \left( 1 - \frac{R_2 R_f}{R_g R_1} \right)$$

$$A_d = \frac{R_g}{R_2 + R_g} \left( 1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1} \right)$$

To make  $A_{cm} = 0$ , we need

$$1 = \frac{R_2 R_f}{R_g R_1}$$

$$\frac{R_f}{R_1} = \frac{R_g}{R_2}$$

at which

$$\begin{aligned} A_d &= \frac{R_g}{R_2 + R_g} \left( 1 + \frac{R_2 + 2R_g}{R_g} \frac{R_f}{R_1} \right) \\ &= \frac{R_g}{R_2 + R_g} \left( 1 + \frac{R_2 + 2R_g}{R_2} \right) \\ &= \frac{R_g}{R_2} \end{aligned}$$

Differential input resistance:

$$i_d = \frac{v_+}{R_g} = \frac{v_2 - v_+}{R_2} = \frac{v_+ - v_2 + v_d}{R_1}$$

$$R_2 v_+ = R_g v_2 - R_g v_+$$

$$R_1 v_2 - R_1 v_+ = R_2 v_+ - R_2 v_2 + R_2 v_d$$

$$v_2 = \left( \frac{R_2 + R_g}{R_g} \right) v_+$$

$$(R_1 + R_2)(v_2 - v_+) = R_2 v_d$$

$$v_+ = \frac{R_g}{R_1 + R_2} v_d$$

$$i_d = \frac{1}{R_1 + R_2} v_d$$

$$R_d = \frac{v_d}{i_d} = R_1 + R_2$$

□

## x Comparator

A comparator is an op amp with  $v_+$  and  $v_-$  connected to the voltage sources to be compared, in which  $v_o$  is high when  $v_+ > v_-$  and low when  $v_+ < v_-$ .

## xi inverting integrator

An inverting integrator consists of an op amp where input voltage  $v_i$  is connected to  $v_-$  through an input resistor  $R_i$ ,  $v_+$  is connected to ground, and  $v_o$  is fed back to  $v_-$  through a feedback capacitor  $C_f$ .

Time domain:

$$i_i = \frac{v_i}{R_i} = i_f = -C_f \frac{dv_o}{dt}$$
$$v_o = v_o(0^-) - \int_0^t \frac{v_i(\tau)}{R_i C_f} d\tau$$

Laplace domain:

$$i_i = \frac{v_i}{R_i} = i_f = -C_f s v_o + C_f v_o(0^-)$$
$$v_o = \frac{v_o(0^-)}{s} - \frac{v_i}{R_i C_f s}$$

## xii inverting differentiator

An inverting differentiator consists of an op amp where input voltage  $v_i$  is connected to  $v_-$  through an input capacitor  $C_i$ ,  $v_+$  is connected to ground, and  $v_o$  is fed back to  $v_-$  through a feedback resistor  $R_f$ .

Time domain:

$$i_i = C_i \frac{dv_i}{dt} = i_f = \frac{-v_o}{R_f}$$
$$v_o = R_f C_i \left( v_i(0^-) - \int_0^t v_i(\tau) d\tau \right)$$

Laplace domain:

$$i_i = C_i s v_i - C_i v_i(0^-) = i_f = \frac{-v_o}{R_f}$$
$$v_o = R_f C_i (v_i(0^-) - s v_i)$$

## xiii Active filter

Given a (passive or active) filter  $v_f$  of an input  $v_s$  and an inverting amplifier, non-inverting amplifier, or unit gain buffer enforcing  $v_o = K v_i$ , by connecting the ground and  $v_i$  of the amplifier to the two terminals of the  $v_f$ , the two terminals ground and  $v_o$  become an active filter of  $v_s$  enforcing  $v_o = K v_f$ .